

REVIEW

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Priority of nutrition and exercise in depression management: triangulating mini-review of past and recent evidence with clinical practice guidelines

Shannon Rogers¹ and Elizabeth Dean^{2*}

Abstract

Background Chronic low-grade systemic inflammation (CLGSI) is implicated in depression and its amelioration. Pro-inflammatory nutrition and inactivity are associated with CLGSI. We triangulated the degree to which recent literature on anti-inflammatory nutrition and physical activity/physical exercise (PA/PE) corroborates the extant body of knowledge related to depression management and is reflected in the guidelines of leading mental health domains (national/international mental health associations/bodies).

Methods We used mini-review methodology. The search was narrowed to recent state-of-the-art literature (2024) in PubMed, on diet and exercise specifications in depression management. Then, we compared recommendations of aforementioned domains regarding diet and PA/PE in depression management.

Results Of 73 nutrition source studies, 50 (68%) focused on anti-inflammatory whole food, plant-based (WFPB) nutrition to manage depression; 4 (5%) on reducing consumption of animal-sourced foods; 16 (23%) focused on the effect of ultra-processed food (UPF) and its role in depression and its avoidance vis-à-vis its anti-depressant effects. Of the 55 PA/PE source studies, 49 (89%) focused on the effects of aerobic exercise; 14 (29%) described specific parameters to achieve an anti-depressant effect and 35 (71%) were non-specific. Twelve (22%) studies focused on resistance muscle training; 2 (17%) that reported specific training parameters and 10 (83%) that were non-specific. Nine domains were identified with established depression management guidelines: Australia/New Zealand; Canada; Europe (Belgium, Scotland, Spain); United Kingdom; United States; WFSBP and ASLM; and World Health Organization. Regarding nutrition, 5 (55%) domains recommended WFPB nutrition; 4 (44%) reduced animal-sourced foods; and 3 (33%), avoidance of UPF. With respect to sedentarism, 3 (33%) domains recommended reduced prolonged sitting. Eight domains (89%) mentioned aerobic exercise; 3 (33%) resistance training. Three domains mentioned aerobic exercise non-specifically; 5 (56%) made specific recommendations. Three domains mentioned resistance muscle training; 2 (22%) made non-specific recommendations and 1 (11%) made specific recommendations.

*Correspondence:
Elizabeth Dean
elizabeth.dean@ubc.ca

Full list of author information is available at the end of the article



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Conclusions Disparities that exist in leading depression management guidelines vis-à-vis inclusion of evidence-informed nutrition and PA/PE recommendations, warrant reconciliation. Evidence supporting anti-depressant WFPB nutrition and limiting pro-inflammatory animal-sourced food and UPF and supporting anti-inflammatory aerobic exercise and resistance training warrants being translated into national/international depression management guidelines as consistently as recommendations for pharmacotherapy and psychotherapy.

Keywords Anti-depressant, Anti-inflammatory, Diet, Nutrition, Physical activity; physical exercise

Background

Depression is the most common mental health condition affecting people worldwide and presents primarily or secondarily in combination with other pathologies [1, 2]. The Diagnostic and Statistical Manual of Mental Disorders describes multiple depression diagnoses including Depression; Depressive disorder due to another medical condition; Depressive disorder due to cerebrovascular disease; Disruptive mood dysregulation disorder; Major depressive disorder; Major depressive episodes and others [3]. The economic burden of these diseases is substantial and unsustainable [4, 5].

Contemporary management of disabling depression includes pharmacotherapy (medication), psychotherapy, and lifestyle modification [6]. Notwithstanding the importance of adequate sleep; manageable stress; and social connections; diet, and sedentarism and physical activity/physical exercise (PA/PE) independently and combined have been documented to contribute to depression bidirectionally in both positive and negative directions [7–16].

Chronic systemic low-grade systemic inflammation (CSLGI) is a common denominator of leading non-communicable diseases (NCDs) including depression [17–26]. Further, CSLGI is associated with the western lifestyle, i.e., consumption of the ubiquitous standard western diet and being sedentary and inactive [27–32]. The standard western diet has been well documented to be nutrient and fiber deficient (e.g., refined, grains, animal-sourced foods, and sugary foods and drinks), consistent with a pro-inflammatory state [33]. This ubiquitous eating pattern has been reported to be pro-oxidative and pro-inflammatory, the consequence of several characteristics that are highly deleterious to human health, e.g., high glycemic load; high fatty acid composition; poor macronutrient composition and quality; low micronutrient density; increased renal acid load and altered acid-base balance; reversal of the potassium-sodium ratio (to high sodium and low potassium); and pathologically low fiber content (characterized by low consumption of legumes, whole grains, vegetables and fruit) [34–37]. As a corollary, nutrient-dense, whole food, plant-based (WFPB) nutrition and regular moderate-intense PA/PE are anti-inflammatory [38–44], and have been reported to improve mood and reduce depression [19, 20, 38–49].

Diet

Various dietary practices and their association with mental health have been studied. Over the past several decades, for example, the consumption of ultra-processed food (UPF) has become prevalent and increasingly a feature of the standard western diet [50]; its reduced consumption associated with improved mental health and reduced depressive symptoms [11, 51–55]. Other dietary regimens have been studied with respect to managing depression, often, if any, are associated with short-term remission including meat consumption that is pro-inflammatory [56–61], ketogenic diet [62–64], and low carbohydrate diet [65–67], and reduced sugar [68, 69].

Physical activity/physical exercise

With respect to being physical inactive, sedentarism is a natural mental health depressant [6]. Thus, the avoidance of prolonged periods of sitting is protective from low mood states and depression. With respect to PA/PE and structured PE in the management of depression, a range of non-traditional alternative physical activities have been studied including tai chi [70, 71], dancing [72], gardening [73], mindfulness [74, 75], qi gong [76], mind body medicine [77], forest bathing [78], Pilates [79], and yoga [80].

Diet and PA/PE combined

The anti-depressive efficacy of optimal nutrition and PA/PE with respect to not only reducing depressive symptoms but also reducing or eliminating the need for medication, is compelling, in both the interest of public health and reducing the economic burden. The potential of healthy nutrition to reduce depressive symptoms may reflect the phytonutrient density associated with fruit and vegetables [81–83]. Similarly, PA/PE appears equally effective in reducing depressive symptoms for individuals with co-morbidities as well as without, and with varying levels of baseline depression [18, 84]. PA/PE has been reported to be 1.5 times more effective at reducing mild-to-moderate symptoms of depression, psychological stress, and anxiety than medication or cognitive behavior therapy [84].

Aim

The overarching aim of this study was to triangulate recent literature with the extant evidence on the roles of anti-inflammatory diet and PA/PE in the management of depression and the clinical practice guidelines for depression. Although based on observation, we predicted a discrepancy between the evidence and practice, the extent of that discrepancy was unclear.

Methods

We conducted a mini-review of recent state-of-the-art evidence to provide a benchmark in the field with respect to specific anti-depressant, anti-inflammatory nutrition and PA/PE recommendations for depression management. First, we scoped the literature to identify diet and PA/PE parameters for potential modification of depression over a recent timeframe (2024) as a means of corroborating the state-of-the-art vis-à-vis the extant body of knowledge; second, to elucidate the degree to which clinical practice guidelines for the management of depression from recognized domains (i.e., national and international mental health bodies that have published formal clinical practice guidelines) include both dietary nutrition and PA/PE recommendations; and third, to establish the degree to which such recommendations reflect the evidence.

Diet and PA/PE parameters for modification of depression

We used established procedures for conducting our triangulating mini-review that were relevant to our topic over one recent year, 2024, in order to corroborate the current state-of-the-art [85]. We chose one recent year given its contemporary relevance, and to establish that recent literature confirms and builds upon the body of extant evidence. We searched one representative, comprehensive database (PubMed) for related peer-reviewed literature. We included peer-reviewed studies using a range of experimental designs, evidence syntheses, epidemiological evidence, and reviews. Search terms included ‘depression’ or ‘depressive disorder’ or ‘depressive symptoms’ or ‘major depressive disorder’) AND (‘exercise’ or ‘physical fitness’ or ‘physical activity’) OR (‘diet’ or ‘nutrition’).

To simplify the associations between diet and PA/PE and depression, we chose to focus on primary depression as much as possible, i.e., without acknowledged co-morbidities. Articles that were excluded were those in the non-English literature (with one exception, Spain) and those related to secondary depression (e.g., associated with a chronic condition), substance abuse and addiction, non-human studies, isolated nutrient supplementation, study protocols, pediatrics, perinatal mothers, athletes, exercise-based gaming, living microbe intake, multifaceted approaches (those that included a range of

interventions other than diet and exercise, e.g., psychological, group work, meditation), COVID-19, emotional eating, food preferences of those who are depressed, food insecurity, those articles that focused on both nutrition and PA/PE combined without differentiation of their effects, and topics for continuing medical education.

Anti-depressant, anti-inflammatory diet and PA/PE recommendations in guidelines for depression management

With independent librarian assistance, we scoped established domains, i.e., organization and national/international bodies, that are dedicated to the needs and management of individuals with depression and published some form of clinical practice guidelines. We accessed the most recent versions of clinical practice guidelines and recommendations for the management of the needs of individuals with depression with special reference to dietary and PA/PE recommendations. We restricted our search to the following English-speaking industrialized countries: Australia and New Zealand, Canada, Europe (Belgium, Scotland, and Spain); United Kingdom; United States; and international collaborative guidelines including the World Health Organization. Recommendations from these nine domains associated with affluent regions, often have a ripple effect and influence guidelines and position papers published in low-middle income countries, lending further support for reviewing them.

Data extraction and tabulation were conducted by the investigators and cross-checked for accuracy. We extracted and enumerated the data based on categories of domain; whether diet and PA/PE recommendations were specified; and, if so, what these were specifically, verbatim.

Results

Our search strategy yielded 73 source studies related to nutrition and its effect on depression. These, their findings, and their implications vis-à-vis dietary recommendations are shown in Supplementary Table (1). The 55 source studies that were related to PA/PE, their findings, and their implications vis-à-vis PA/PE recommendations are shown in Supplementary Table (2). An evidence-based summary of diet and physical activity/physical exercise parameters and recommendations for prevention and modification of depression appear in the Figure 1.

Dietary parameters for modification of depression

Of 73 nutrition source studies (Supplementary Table 1), 50 (68%) focused on anti-inflammatory WFPB nutrition to manage depression; 4 (5%) on reducing consumption of animal-sourced foods; 16 (23%) focused on the effect

Panel A***Assessment of Diet*****Anti-inflammatory constituents**

- Whole food
- Vegetables, fruits, legumes, whole grains
- Nuts and seeds
- Ultra-processed food

Pro-inflammatory constituents

- Animal-sourced foods (meat, dairy, eggs) including processed meats
- Ultra-processed foods
- Fat, sugar and salt

Anti-inflammatory Dietary Recommendations

- Diet rich in plant-based foods (Note: due to sanitation requirements, vitamin B12 is no longer widely accessible through food and water supplies, thus, the more plant-based a person's diet, there is a greater need for vitamin B12 enriched foods or supplementation)
- No animal-sourced foods, if any, to reduce endotoxin load associated with saturated fat
- No ultra-processed foods that are high in fat, sugar, and salt, and are engineered with cheap commodity crops and hormone-disrupting additives
- No sugary drinks including soft drinks; preferably whole fruits rather than juices

Panel B***Assessment of Sitting and Physical Activity/Physical Exercise***

- Sitting time
- Physical activity
- Structured exercise
- Sport participation

Anti-inflammatory Recommendations for Reducing Sedentarism and Increasing Physical Activity/Physical Exercise

- Reduce prolonged sitting time; get up and walk around every 20-30 minutes
- Engage in regular daily physical activity and walking throughout the day; outdoors in natural settings as much as possible
- Engage in structured physical exercise:
 - Aerobic exercise: walking/jogging/swimming to get exertion into the moderately-intense range, 20-40 minutes, 3-5 times/week
 - Resistance muscle training: weights, bands, machines to get intensity into the moderate range, typically 3 sets of 10 repetitions. 2-4 times/week

Fig. 1 Evidence-based summary of diet (Panel A) and physical activity/physical exercise (Panel B) parameters and recommendations for prevention and modification of depression

of ultra-processed food (UPF) and its role in depression and its avoidance vis-à-vis its anti-depressant effects. Note, that some studies focused both on plant-based nutrition and avoidance of UPF.

PA/PE parameters for modification of depression

Of the 55 PA/PE source studies (Supplementary Table 2), 49 (89%) focused on the effects of aerobic exercise; 14 (29%) described specific parameters to achieve an anti-depressant effect and 35 (71%) were non-specific. Twelve (22%) studies focused on resistance muscle training; 2 (17%) that reported specific training parameters and 10 (83%) that were non-specific. Note, that some studies focused both on aerobic and resistance muscle training.

Diet and PA/PE guidelines for the management of depression published by established domains

Nine domains were identified with established guidelines: Australia/New Zealand; Canada; Europe (Belgium, Scotland, Spain); United Kingdom; United States; and World Federation of Societies for Biological Psychiatry (WFSBP) and Australasian Society of Lifestyle Medicine (ASLM); and World Health Organization. The diet and exercise guidelines for the management of depression from these established sources are summarized in Table 1.

With respect to nutrition (Table 2), 5 (55%) domains recommended WFPB nutrition; 4 (44%) reduced animal-sourced foods; and 3 (33%), avoidance of UPF. With respect to PA/PE (Table 2), 3 (33%) domains recommended reduced prolonged sitting (sedentarism). Eight domains (89%) mentioned aerobic exercise; 3 (33%) resistance training. Three domains mentioned aerobic exercise non-specifically; 5 (56%) made specific recommendations. Three domains mentioned resistance muscle training; 2 (22%) made non-specific recommendations and 1 (11%) made specific recommendations.

Discussion

Pharmacotherapy and psychotherapy remain the hallmark of contemporary depression treatment, either alone or with other treatment strategies. The findings of this mini-review however confirmed that the emphasis on anti-inflammatory nutrition and PA/PE in practice guidelines for depression, pales in comparison to the emphasis on pharmacotherapy and psychotherapy, which is not consistent with the weight of the evidence. Although nutrition and PA/PE interventions may have limitations and their implementation warrants being gauged to the needs of each individual, this is also true for the administration of pharmacotherapy and psychotherapy.

Consistent with a biomedical model, contemporary guidelines disproportionately weight psychotherapy and psychopharmacology [86]. As important as these

interventions may have been shown to be, their emphasis tends to overshadow the contributions of the contemporary standard western diet and PA/PE practices that can contribute to and perpetuate depressive symptoms. Although the association of nutrition and mental health, for example, has been recognized for several decades giving rise to a psychiatric specialty of nutrition psychiatry, clinical practice guidelines for the management of psychiatric conditions have yet to fully integrate this body of knowledge into standard practice [87].

Diet and PA/PE parameters for modification of depression

The optimal parameters for diet and PA/PE in the modification of depression are detailed below and are summarized in the Figure.

Diet.

With respect to diet, despite general agreement in the extant literature favoring WFPB nutrition in the management of depression, there were some anomalous findings in the recent literature that warrant explanation. For example, Apostolakopoulou and colleagues [8] concluded there were inconsistent reports regarding the anti-depressant effects of vegetarian diets. However, other than the confounded study by Saintila and colleagues that is described below [88], two source studies that defined vegetarian diet (e.g., consumption of fruits, vegetables, and whole grains) reported an inverse relationship with depression [89, 90] which supports earlier studies. Other anomalies, were two Asia studies that reported an inverse relationship between the consumption of fish and eggs, foods that are generally regarded as pro-inflammatory, and depressive symptoms [91, 92]. Compared with the standard western diet, the traditional Asian diets are more WFPB with lower fat, sugar and salt. One study of an American cohort reported that animal-sourced protein as well as anti-inflammatory vegetable protein and high-quality carbohydrate have a negative association with depression [93]. WFPB nutrition by definition is associated with high quality, unrefined carbohydrate content. Individual studies confirm such high-quality carbohydrate consumption is associated with low depression risk [18, 94] and low-quality, refined and highly processed carbohydrate is associated with depressive symptoms [95]. One study reported a higher caloric ratio of carbohydrate intake was associated with greater depression however the source of carbohydrate was not differentiated (complex or refined) [96], i.e., anti-inflammatory complex carbohydrates characteristic of WFPB nutrition, vs. pro-inflammatory simple carbohydrates characteristic of the standard western diet. Cheng and colleagues [97] reported that moderate carbohydrate (undefined) and higher protein consumption (plant or animal sources undifferentiated) were depression protective.

Table 1 Diet and physical activity/physical exercise parameters for potential modification of depression recommended by nine established domains (associations and national and international health reports)

Country/ Domain	Association and National/ International Health Reports	Dietary recommendations and Co-morbidity implications	Exercise recommendations and Co-morbidity implications
1 Australia and New Zealand	Malhi et al. The 2020 Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. Aust N Z J Psychiatry. 2021;55(1):7–117	The evidence supporting the ‘Mediterranean diet’ characterised by high vegetable, fruit, fish and grain components, with low animal fat components, has increased The content of diet has an impact upon mood disorder symptoms, psychological well-being and overall health Ample amounts of fruits, vegetables (including potatoes), breads (particularly whole grain), nuts and seeds Fresh rather than processed food Fresh fruits as preferred desserts Dairy, poultry and fish in moderate quantities Red meats in small amounts Reduce saturated fats Olive oil (or other vegetable oil if preferred) as prime source of dietary fat Alcohol in modest quantities: maximum 1–2 standard drinks/day for males and 1 for females Methylfolates, probiotics and Omega-3 fatty acids may be helpful	Regular exercise is associated with improved quality of life, and antidepressant effects in depressive and bipolar disorders Exercise – particularly aerobic exercise – should be encouraged in all patients, not least because of its general health benefits Although the optimal amount and type of exercise is yet to be determined, we do know that health benefits require aerobic activity to be vigorous (enough to require concentrated effort) and regular (at least two to three times weekly) Resistance-based exercise is also beneficial in the management of depressive symptoms, and a combination of aerobic and resistance exercises is probably optimal Helping a patient build an exercise plan: Inform the patient that exercise is a helpful adjunctive treatment for depressive disorders because it improves brain function, offers pleasurable experiences, improves self-esteem, enhances sleep quality and increases the opportunities for interactions with other people Daily exercise is preferred over intermittent activity Aerobic (e.g., walking, running, cycling) and resistance exercise (e.g., using weights) are both effective for depressive disorders, and the combination may be better than either alone Some patients may find exercising with other people more motivating, but this is not universally the case Choose forms of exercise according to personal preference, remembering that previous preferences for exercise activities may help with this choice Ideally exercise should be pleasurable and require some effort, but not be exhausting or painful Documenting an exercise plan is helpful, along with regular reviews of progress and difficulties Some patients with depressive disorders struggle to find the ‘right exercise’ for them. Suggestions based upon their past activities and a review of options (e.g., start with a walk every day, perhaps with another person or pet) may be helpful Patients will benefit from building on their exercise plans to make these activities part of their daily lives; this may start as ‘treatment’ but could progress to becoming a ‘lifestyle’

Table 1 (continued)

Country/ Domain	Association and National/ International Health Reports	Dietary recommendations and Co-morbidity implications	Exercise recommendations and Co-morbidity implications
2 Canada	Lam et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) 2023 Update on Clinical Guidelines for Management of Major Depressive Disorder in Adults (CANMAT) 2023. <i>Can J Psychiatry</i> . 2024 Sep;69(9):641–687	Adjunctive healthy diet (varied diet with high content of fruit, vegetables, and fiber, and low content of saturated fat and carbohydrates)	Substantial evidence for the efficacy of exercise as a treatment for major depressive disorders adjunctive exercise is superior to no treatment and treatment as usual conditions in major depressive disorders, with medium to large effects Benefits in reducing depressive symptoms with the supervised aerobic activity of moderate to high intensity in adolescents and young adults, with low to moderate intensity exercise in midlife and older women, and with high-intensity interval training in adults First line: Supervised exercise (i.e., low to moderate intensity, for 30 to 40 min at a time, 3 to 4 times a wk, for a minimum of 9 wks) as a first-line monotherapy for mild depression and a second-line adjunctive treatment for moderate severity illness Supervised exercise (low to moderate intensity, for 30 to 40 min at a time, 3 to 4x/wk, for a minimum of 9 wks) for major depressive disorders of mild severity
3 Europe: Belgium	Medina et al. European clinical practice guidelines for depression in adults: Are they good enough? <i>J Affect Disord</i> . 2020 Feb 15;263:382–385. doi: https://doi.org/10.1016/j.jad.2019.12.005 . Epub 2019 Dec 5. Belgium Karyotaki et al. (2014). <i>The long-term efficacy of psychotherapy, alone or in combination with antidepressants: The treatment of adult major depression</i> (KCE Report 230 C). Belgian Health Care Knowledge Centre (KCE)	None stated	None stated
4 Scotland	Scotland/Scottish Intercollegiate Guidelines Network. (2010). <i>Non-pharmaceutical management of depression in adults</i> (SIGN Guideline No. 114). Healthcare Improvement Scotland	None stated	Structured exercise may be considered as a treatment option for patients with depression Exercise parameters are based on a limited number of studies including the following minimum requirements: three sessions/wk; 30–40 min duration; and a total energy expenditure of 17 kcal/kg/wk Other recommendations: 5 sessions/wk, 30 min at a lower intensity with similar total energy expenditure Required intensity of exercise correlated to energy expenditure of 70–80% of heart rate reserve Co-morbidities not addressed

Table 1 (continued)

Country/ Domain	Association and National/ International Health Reports	Dietary recommendations and Co-morbidity implications	Exercise recommendations and Co-morbidity implications
5 SpainUnitedKingdom*(*see item 6)	<i>Spain</i> Working Group of the Clinical Practice Guideline on the Management of Depression in Adults. (2014). <i>Clinical practice guideline on the management of major depression in adults</i> . Ministry of Health, Social Services and Equality, Government of Spain <i>United Kingdom*</i> Guidelines from the United Kingdom in the Medina article were based on the 2010 NICE report; there are excluded here as these guidelines have been updated in 2022 and appear below	(From translation)None stated	(From translation)Limit sedentarismPhysical activity should be considered a part of treatment in adults with episodic depressionIn moderate depression, physical activity should be considered as a compliment to treatment either with antidepressant medications or structured psychological treatmentsA program of structured and supervised physical exercise for 30–60 min, 3 times/wk for ≥ 10–12 wksCo-morbidities not addressed
6 United Kingdom	National Institute of Health and Health and Clinical Excellence (NICE) Nice.org.uk/guidance/ng222 (2022)	Eating a healthy, balanced diet is an important part of maintaining good health, and can help you feel your best • eat at least 5 portions of a variety of fruit and vegetables every day (see 5 A Day) • base meals on higher fiber starchy foods like potatoes, bread, rice or pasta • have some dairy or dairy alternatives (such as soya drinks) • eat some beans, pulses, fish, eggs, meat and other protein (choose lean cuts of meat and skinless poultry whenever possible to cut down on fat; and less red and processed meat like bacon, ham and sausages) • choose unsaturated oils and spreads, and eat them in small amounts • drink plenty of fluids (at least 6 to 8 glasses a day) • Healthy Eating as a Vegan: this section is consistent with an anti-inflammatory diet	General Physical Activity Guidelines for Adults Aged 19 to 64 (also those 65 y and over) Provisos re other superimposed conditions • do strengthening activities that work all the major muscle groups (legs, hips, back, abdomen, chest, shoulders and arms) on at least 2 days/wk • do at least 150 min of moderate intensity activity a week or 75 min of vigorous intensity activity a week • spread exercise evenly over 4 to 5 days a week, or every day • reduce time spent sitting or lying down and break up long periods of not moving with some activity Group exercise with a trainer undertaking moderate intensity aerobic exercise Usually consists of ≥ 1 session/wk for 10 wks

Table 1 (continued)

Country/ Domain	Association and National/ International Health Reports	Dietary recommendations and Co-morbidity implications	Exercise recommendations and Co-morbidity implications
7 United States	Guideline Development Panel for the Treatment of Depressive Disorders. Summary of the Clinical Practice Guideline for the Treatment of Depression Across Three Age Cohorts. American Psychological Association. <i>Am Psychol.</i> 2022;77(6):770–780	Nutrition was not a primary focus of the recommendations Guidelines acknowledge some evidence supporting dietary interventions, including the Mediterranean diet A diet rich in vegetables, fruits, fish, and grains, with low animal fats, appears beneficial for depressive disorders Olive oil is recommended as the primary source of dietary fat Red meat intake should be minimized Socioeconomic challenges must be considered when recommending dietary changes A dietitian or nutritionist may be required to support long-term adherence to dietary modifications Co-morbidities Diet and exercise interventions should be adapted for those with metabolic or cardiovascular conditions (e.g., obesity, hypertension) Monitoring weight, blood glucose, and lipid levels is essential in managing depressive patients with metabolic syndrome Depression may reduce motivation to engage in lifestyle changes, requiring structured support Family and community involvement in dietary and exercise programs can enhance adherence	Exercise and diet are beneficial for mild depressive symptoms Regular physical activity is strongly encouraged, with 150 min/wk of moderate exercise are recommended to help alleviate depressive symptoms Supervised exercise programs provide better adherence and outcomes than unsupervised regimens General physical activity Regular exercise is emphasized as a key intervention for depression management Daily exercise is preferable, and both aerobic (walking, running, cycling) and resistance training (weights) are effective Moderate and vigorous exercise (150–860 METs-min/wk) effectively reduces depression, anxiety, and stress. Exercise enhances sleep quality, self-esteem, cognitive function, and neurotransmitter regulation Group exercise programs for depression Typically include moderate-intensity aerobic exercise for more than one session/wk for 10 wks 8 participants per group is typical Supervised programs (8–12 wks, 3–4 sessions/wk) improve functioning Exercise type includes walking, jogging, yoga, and strength training significantly reduce depression symptoms

Table 1 (continued)

Country/ Domain	Association and National/ International Health Reports	Dietary recommendations and Co-morbidity implications	Exercise recommendations and Co-morbidity implications
8 World Federation of Societies for Biological Psychiatry (WFSBP) and Australasian Society of Lifestyle Medicine (ASLM)	Marx et al. Clinical guidelines for the use of lifestyle-based mental health care in major depressive disorder: World Federation of Societies for Biological Psychiatry (WFSBP) and Australasian Society of Lifestyle Medicine (ASLM) taskforce. <i>World J Biol Psychiatry</i> . 2023 Jun;24(5):333–386	Encourage adherence to nutrient-dense, minimally processed dietary patterns, such as the Mediterranean diet Where required and available, refer to a trained dietician Increase consumption of fruits, vegetables, legumes, whole grains, nuts, seeds, herbs and spices as tolerated Include a high consumption of foods rich in omega-3 polyunsaturated fatty acids and fiber Limit intake of ultra-processed foods and discretionary foods, and replace ultra-processed foods with minimally processed nutritious foods Consume red meat in moderation and opt for lean sources rather than processed and/or fatty cuts, considering the individual's cultural-religious background Include extra virgin olive oil as the main source of cooking and added oil Consume the daily recommended water intake Avoid excessive alcohol consumption	Inquire about and encourage individuals to engage in modes of physical activity that they enjoy and at a frequency and intensity that they can sustain Discuss the use of supervised physical activity options such as group classes, use of personal trainers, and team sports as these may improve adherence in some individuals Engaging with an accredited exercise physiologist may be warranted especially where physical co-morbidities exist for individuals with major depressive disorders (e.g., heart conditions, COVID-19) to overcome barriers to participating in physical activity Pairing exercise with enjoyable activities can increase motivation e.g., listening to music, socializing, or exercising with a partner, watching television, or exercising in a pleasant environment Positive mental, physical, and social experiences acknowledged during physical activity (in real-time as opposed to retrospectively) can guide attention and increase motivation Encourage exercise routines that are feasible to implement most days rather than sporadically and that are effortful but not too difficult, exhausting, or painful. Examples may include commencing with initially low-moderate intensity exercise (e.g., walking/cycling short distances) rather than commencing with high intensity exercise routines (e.g., sprints, long distance running) Where feasible, work with individuals to gradually incorporate bouts of higher intensity exercise to gain maximal antidepressant benefits Provide examples of ways to reduce sedentary behavior as well as improving physical activity (e.g., digital apps and push notifications, reminders, standing/walking meetings, environmental modifications such as standing desks)
9 World Health Organization	mhGAP Intervention Guide Mental Health Gap Action Programme Version 2.0 for mental, neurological and substance use disorders in non-specialized health settings. 2016 https://www.who.int/publications/item/9789241549790	Mention of assessing inappropriate diet along with other health-harming lifestyle behaviors and chronic disease 'Try to eat regularly despite changes in appetite'	Mention of assessing physical inactivity along with other health-harming lifestyle behaviors and chronic disease 'Try to be as physically active as possible'

Two source articles failed to consider confounding sociodemographic variables, thus warrant particular mention to explain their exclusion from analysis. First, contrary to other evidence, Marche and colleagues reported that high protein consumption is associated with lower risk of depression [59]. In that study, the ratio of family income-to-poverty was lower ($p < 0.001$) for those study participants who were depressed. Individuals and families in lower socio-economic groups have

been documented to eat less well and have nutritional deficiencies and are more likely to experience depression related to socioeconomic status and health issues [98, 99]. Second, Saintila and colleagues [88], reported that vegetarian diets are associated with increased symptoms of depression. Comparable to sociodemographic confounding variables in the study by Marche and colleagues (60), the vegetarians in the study by Saintila and colleagues were Peruvians with lower-socioeconomic

Table 2 Summary of the nine domains that included evidence-based depression management guidelines related to anti-inflammatory nutrition and physical activity/physical exercise

	Country/Domain	Whole food, plant based diet	Reduced animal-sourced foods	Avoid ultra-processed food	Reduced sedentarism	Aerobic	Resistance muscle training
1	Australia and New Zealand	✓	✓			✓ Non-specific	✓ Non-specific
2	Canada	✓	✓	✓		✓ Specific	
3	Europe: Belgium					✓ Specific	
4	Europe: Scotland					✓ Specific	
5	Europe: Spain				✓	✓ Specific	
6	United Kingdom	✓	But does acknowledge vegan nutrition	✓	✓	✓ Specific	✓ Specific
7	United States	✓	✓			✓ Specific	✓ Non-specific
8	World federation of societies for biological psychiatry (WFSBP) and Australasian society of lifestyle Medicine (ASLM)	✓	✓	✓	✓	✓ Non-specific	
9	World health organization					✓ Non-specific	

status, compared with the non-vegetarians ($p < 0.001$) [88]. Those in low-socioeconomic groups have been documented to be more likely to consume nutrient-poor diets irrespective of whether they report being vegetarian or not. The United Nation's Food and Agriculture Organization reported Peru has become the most food-insecure country in South America affecting over 50% of the population [100].

Diet recommendations

Superimposed on the complexity of managing the needs of people who are depressed, is their reduced motivation to eat well including altered food preferences that are often processed and less healthy [101, 102], or to move and exercise [103, 104]. Thus, a detailed initial assessment is warranted to establish the patient's current nutrition and exercise practices, as well as their histories, and readiness to consider such lifestyle changes in addition to their readiness to take medication or be referred for psychotherapy [105].

Compared with pharmacotherapy and psychotherapy, contemporary dietary and PA/PE recommendations are often categorized as adjunctive rather than among primary interventions when included in depression management approaches [86]. This is curious given that anti-inflammatory nutrition and PA/PE interventions may not only offset depressive symptoms but also can avert chronic NCDs that are often associated with depression [106]. These lifestyle interventions are also less invasive, have no adverse side-effects, are

inexpensive, and are associated with greater assurance of life-long health overall.

Key findings from review of 2024 literature on diet and depression (Supplementary Table 1) corroborate extant literature described in the introduction. Inflammation contributes to depression and is also a consequence of depression [107]. Systemic inflammation influences neuroendocrine function and neurotransmitter metabolism, as well as synaptic efficiency. Proinflammatory signaling pathways also contribute to the pathogenesis of depression [29, 108]. High-fiber diets that are nutrient dense reduce inflammation by modulating the gut microbiome, in turn, influencing mood via the gut-brain axis [47, 107]. The gut-brain axis modulates intestinal inflammation [107]. Depressive symptoms have been associated with inflammatory compounds including sucrose, polyunsaturated fats and omega-6 linoleic acids, whereas anti-inflammatory compounds omega-3 (eicosapentaenoic acid) and vitamin C ameliorate depressive symptoms [109].

In one recent study of Australian women, Lee and colleagues concluded that depressive symptoms were reduced as long as the diet was high quality (defined as rich in plant foods) regardless of whether it was plant-based or not, and that a low-quality diet may increase depressive symptoms [110].

Overall mental health outcomes are improved with a diet rich in plant-based foods and less animal-sourced foods [111]. Diets that are higher in plant than animal content and fermented dairy products which encourage

microbial diversity in the gut, can attenuate intestinal inflammation by modulating the gut microbiome [107]. One example is the EAT-Lancet diet primarily plant-based and emphasizes the intake of vegetables, fruits, whole grains and nuts [112]. It contains moderate amounts of seafood and poultry, and significantly limits red meat, and restricts added sugar and saturated fat. This dietary pattern limits animal-sourced foods, emphasizes higher plant content to better modulate the gut microbiome, thereby reduces depression and anxiety [112]. In one study, higher vegetable intake was linked to reduced depression, particularly in women, which is consistent with the literature. Participants reported fewer depressive symptoms with higher egg consumption, a finding that has not been independently corroborated [92].

Limiting meat consumption has been reported to reduce inflammation given animal protein is pro-inflammatory which does not support mental health [107]. However, some evidence suggests that protein sources including unprocessed red meat and poultry, are negatively correlated with the risk of depression as are the inclusion of dairy and nuts in the diet [93]. In one study, consuming chicken meat, fish, cereals, cheese as well as fruit, has been reported to be protective for depression and suggested that particularly in elderly populations, animal-sourced foods may reduce depression [58]. Overall, however, evidence strongly supports a plant-based diet such as the Mediterranean diet [112] for optimal mental health including reducing depression symptoms [88, 111]. Methodologically, these discrepancies warrant further investigation particularly in light of the well-documented health-harming effects of animal-sourced foods on human health [36]. Furthermore, endotoxins associated with meat produce symptoms resembling depression [113]. Bacterial toxins in the circulation that follows a meal of animal products results in inflammation and stiffened arteries, potentially exacerbated by the presence of saturated animal fat [114–116].

Much has been written about plant-based nutrition for maximal health and reduced disease risk including veganism [36]. Joy has providing evidence that eating animal-sourced foods is socially and culturally determined, rather than these foods being needed for maximal human nutrition [117]. Consistent with this (with the possible exception of the United Kingdom), is that 3 (33%) of the recommendations from the nine domains recommended a reduction of animal-sourced foods. Several leading global nutrition and dietetic associations have concurred globally that vegan nutrition is healthy across the life cycle including pregnant and lactating mothers [118]. However, largely due to sanitation requirements and contemporary agricultural practices, vitamin B12 is no longer widely accessible through food and water supplies.

Thus, patients who are or become vegan, vitamin B12 enriched foods or supplementation is essential [36].

With respect to UPF, its consumption contributes to depression through inflammation-related mechanisms [119]. Sugar, for example, is a prime ingredient of processed foods. High-sugar diets are positively associated with depression in adults [44, 107]. Given diet's impact on the gut microbiome, food choices can attenuate intestinal inflammation. High-sugar diets which include soft drinks and alcohol, disrupt the gut microbiota, leading to depression. Limiting the intake of sucrose and fat, basic ingredients of processed foods, may attenuate depressive symptoms by assisting the limbic system's regulation of brain-derived neurotrophic factor [109]. Avoiding a pro-inflammatory diet including UPF such as junk food, fast food, processed meat, and alcoholic and soft drinks, lowers the risk of depression [49, 107].

Key findings from review of 2024 literature on PA/PE and depression (Supplementary Table 2) also corroborate extant literature described in the background of this review. Independently, sedentarism depresses mood and is associated with increased risk of depression in adults and older adult populations [120]. Regular PA/PE prevents depression onset regardless of age, sex or geographical region [121, 122]. PA/PE can be as effective as psychotherapy and antidepressant medication for treating depression even in those with transdiagnostic and subthreshold symptoms [18, 123]. Both aerobic exercise and resistance training are associated with significant therapeutic benefits when used as monotherapy or as adjuncts to medications and psychotherapy [124]. Some forms of physical exercise have been reported to be even more effective than selective serotonin reuptake inhibitors alone [18]. PA/PE can augment the effects of selective serotonin reuptake inhibitors; therefore, a combined exercise and medication approach may offer substantial benefits [18].

PA/PE

The principles and parameters of PA/PE follow that are consistent with an anti-depressant effect.

PA/PE recommendations

Despite the important of PA/PE, periods of prolonged sitting also need to be limited.

Sedentarism is an independent risk factor for multiple conditions including depression as was documented after the COVID-19 pandemic [125]. Thus, first and foremost, a case can be made for recommending reduced periods of prolonged sitting, to individuals who are depressed, as much as possible given the stage and severity of depression. Sitting time can be broken up every 20 to 30 min.

Even minimal amounts of PA/PE can decrease depressive symptoms [123]. Simply, increasing weekly energy

expenditure by engaging in various physical activities including walking, jogging, yoga, and strength training can reduce levels of depressive symptoms [18, 80]. Combined aerobic and resistance training has been reported to reduce depression particularly in those with moderate symptoms [80]. Together, they showed moderate-to-large reductions in depressive symptoms and aerobic exercise was reported to be a particularly effective coping strategy [66].

Any intensity of PA/PE can be effective for reducing depression in a dose-dependent manner, however vigorous or intense exercise had the largest effect in the general adult population [18, 66]. For older adults, moderate intensity PA/PE appears to have the greatest effect on reducing depression until 81 years of age, after which its effect appears more limited [126, 127]. In frail and older adult populations, supervised individual or group interventions were particularly effective [127].

The predicted benefit of various PA/PE prescriptions depended upon co-morbidities, age and baseline levels of depression [18]. Sessions longer than 30 min were more effective in reducing depressive symptoms with weekly accumulation of only 150 min/week of moderate aerobic exercise [128]. Other studies described optimal energy expenditure to decrease symptoms in adults as being 600–1200 METs of light to moderate intensity [129]. However, in older adults, walking was effective in reducing depression in intensities over 250 METs/wk. Exceeding 1000 METs-mins/wk did not demonstrate substantial advantages in symptom reduction [80]. Various dose-response profiles for various physical activities were effective at 320 METs-min/wk and were optimized at 860 METs-min/wk [127]. Interventions ranging between 9 and 24 wks with a minimum duration of 8–12 wks, with supervised PE sessions occurring 385–4 times/wk were associated with the best outcomes [123].

In sum, PA/PE is an effective intervention in depression management and provides a potent alternative or adjunct to pharmacological approaches which are associated with side effects and limited long term efficacy [130]. There are also potential cost benefits to implementing group exercise interventions benefiting patients with depression and health systems [18].

Diet and PA/PE recommendations in depression management guidelines

The contemporary recommendations for managing depression largely reflect the biomedical model with a focus on drugs and psychotherapy and less emphasis on lifestyle interventions known to impact depression including nutrition and PA/PE. Where there is attention to nutrition and PA/PE, recommendations are variable, and not consistently aligned with the evidence. Notably, however, the Canadian revised guidelines note that there

is level one scientific evidence supporting exercise as a first-line lifestyle treatment for major depressive disorder of mild severity [131]. It includes clear and specific prescription parameters based on the evidence, i.e., frequency, intensity, time and type of PA/PE. The Canadian Network for Mood and Anxiety Treatments (CANMAT) also offers second-line lifestyle guidance for PA/PE as adjunctive therapy in moderate severity major depressive episodes. Adjunctive nutrition interventions including the Mediterranean diet and a diet with high fruit, vegetables and fiber, and low saturated fat and carbohydrates are considered third-line treatment recommendation. The inclusion of specific recommendations regarding diet and PA/PE sets this guideline apart from the others as the only clinical practice guideline which represents the science regarding the effects of diet and PA/PE on managing depression [131].

The literature supporting the benefit of WFPB nutrition for optimal mental health including depression as well as physical health (optimal physical health augments mental wellbeing) is established and unequivocal. Although all types of PA and PE appear to benefit mental health, there are dose-dependent benefits as described above. Compared with PA/PE recommendations, dietary recommendations are more uniform across individuals who may be or are depressed. PA/PE recommendations, on the other hand, need to consider type of depression, its presenting symptoms and impact on the patient, co-morbidities, individual preferences, age, and motivation. How best to tailor and target recommendations for diet and PA/PE and long-term adherence, for optimal mental health for a given individual, warrants being established with behavioral strategies including cognitive behavior therapy and motivational interviewing. Optimal nutrition and PA/PE prescription may help to avert the need for medication or, minimally, reduce its need.

Both the American College of Sports Medicine and the Canadian Society for Exercise Physiology recommended PE as an adjunctive treatment for individuals who have depressive disorders [132, 133]. Their specific recommendations are consistent with the literature. These two domains however were not included in those domains that we identified as providing clinical practice guidelines, as their focus was singularly PE. Having said that, the recommendations by the Canadian Society for Exercise Physiology also acknowledged diet and that it should be varied. Specific mention was made of the Mediterranean diet [133].

Degree to which diet and PA/PE recommendations for depression reflect the current evidence base

Of the nine domains, the recommendations of the domains of Australia and New Zealand; United Kingdom; United States, and the WFSBP and ASLM were

most aligned with the science vis-à-vis anti-inflammatory nutrition and PA/PE, with the latter domain being most aligned. These recommendations included WFPB nutrition and reduced animal-sourced foods. Only one domain, the WFSBP, specifically cautioned against consuming UPE, although some of the others inferred it.

Strengths

Depression has been reported to have an associated inflammatory component. To our knowledge, this is the first examination of the congruence between the evidence related to the anti-inflammatory components of nutrition and PA/PE, and established clinical practice guidelines for depression management.

Limitations

Although the science supporting WFPB nutrition and regular moderate-intense aerobic and resistance exercise being unequivocally associated with reduced oxidative stress and CLGSI contributing to depression, their inclusion into depression management regimens is but one aspect of comprehensive depression management, as substance abuse, poor sleep, social connections, and psychological stress need to be considered.

Further, similar to biomedical studies, the degree to which placebo plays a role in studies' findings needs to be addressed. The issue of a placebo effect of exercise has been raised and warrants consideration in future study designs [134, 135]. The issue of placebo also may be a valid issue in relation to dietary practices as well.

Although some studies supported that the consumption of animal-sourced foods were inversely related to depression in some cohorts, this mini-review focused on the benefits of anti-inflammatory nutrition which has been documented to anti-depressant. As described, animal-sourced foods have been documented to be pro-inflammatory [136–138]. With respect to engaging in no exercise vs. anti-inflammatory exercise, no discrepancy has been reported in the literature. Typically, the focus of exercise-related studies is on the parameters of exercise that are associated with the maximal anti-depressant effects.

Our mini-review focused on general non-acute depression management. It is reasonable that the benefits of anti-inflammatory nutrition and PA/PE may benefit individuals with a range of depressive disorders to varying degrees. That said, comparable to the prescription of medication and psychotherapy, the timing of introducing nutrition and exercise interventions at a given stage in the patient's course of depression, is based on a comprehensive assessment and ongoing evaluation. Comparable to other conditions, if depression becomes chronic, there is the potential that addressing lifestyle factors such as

nutrition and PA/PE could prevent such episodes in the future and reduce the need for medication [139].

Finally, with respect to benchmarking the recent literature, we opted to use one representative established database. This was done for expediency to enable investigators to examine each study in sufficient detail to serve as a baseline for comparison with previous literature. That other databases might have yielded other findings is debatable. Also, in future studies and in the clinical practice guidelines that their findings inform, terms and their variants need to be clearly defined, e.g., 'exercise' and 'diet'.

Clinical and research implications and directions

Studies are needed to better prescribe anti-inflammatory diet and PA/PE to individuals with depression, e.g., readiness to introduce lifestyle changes. Individuals who are depressed have been reported to eat poorly and exercise less. Thus, assessing these individuals with respect to how they might most expediently and effectively shift their diets to being more plant-forward, and sit less and be aerobically active is paramount. Then, working with each individual to develop a unique program for each in improving the healthfulness of their lifestyle including anti-depressant, anti-inflammatory WFPB nutrition and PA/PE. Follow-up and provision of support are central to effecting change. Mixed methods research including qualitative research methods may shed more light on aspect to consider in both assessing individuals with depression on their lifestyles, and in prescribing individual lifestyle behavior change. Behavioral strategies including cognitive behavior therapy and motivational interview may have a role in empowering individuals with depression assume a pro-active role in addressing their symptoms.

Given the pervasive benefits of reduced sedentarism, and increased PA/PE on depressive symptoms, clinicians need to judiciously prescribe the appropriate dose of exercise based on the patient's individual attributes and needs in conjunction with other indicated care. This integrated approach holds the promise of maximizing improvements in symptoms for patients with depressive symptoms. Some studies have shown that the benefits of PA/PE can be superior to medication, thus exploiting exercise is less costly and a more empowering alternative for patients [140, 141]. Future studies need to establish those who may benefit from this combination alone, i.e., nutrition and exercise, and for whom their benefit may be augmented with medication. Also, can this combination aid in the de-prescribing of medication in this patient cohort is provocative and warrants investigation. The relative effects of diet and exercise may vary from one individual to another, thus further study may elucidate this distinction. Future studies are also needed to elucidate

the role of individual differences with respect to personal health autonomy and perceived control of their depressive symptoms and their capacity to influence them, e.g., differences impacted by gender, age, culture, and co-morbidity. Mixed-methods research is needed to investigate the barriers and facilitators of diet and PA/PE interventions for depression management from the perspectives of both the clinicians who recommend and support them, and those affected with depression who integrate these interventions into their symptom management.

Finally, despite evidence supporting an anti-inflammatory role underlying the benefit of an anti-oxidant, nutrient-dense, plant-rich diet and PA/PE to mental health, particularly depression, potential other mechanisms that warrant elucidation.

Conclusion

Disparities that exist in leading depression management guidelines vis-à-vis inclusion of evidence-informed nutrition and PA/PE recommendations, warrant reconciliation. Evidence supporting anti-depressant WFPB nutrition and limiting pro-inflammatory animal-sourced food and UPF and supporting anti-inflammatory aerobic exercise and resistance training warrants being translated into national/international depression management guidelines as consistently as recommendations for pharmacotherapy and psychotherapy. We recommend that such integration be undertaken in the next iteration of the depression management guidelines published by the nine domains of interest, as well as new domains.

With attention to evidence-informed nutrition and PA/PE recommendations, not only may depression be prevented or offset in some individuals, but it may empower the person to exert control over their depressive symptoms and help reduce the need for medication. Maximizing personal control and agency is one means of augmenting mental health at a judicious time, such that depression is prevented or ameliorated. This relationship can be bidirectional so identifying the weighting of these factors in each individual presentation is imperative.

Abbreviations

ASLM	Australasian society of lifestyle medicine
CANMAT	Canadian network for mood and anxiety treatments
CLGSI	Chronic low-grade systemic inflammation
NCDs	Non-communicable diseases
PA	Physical activity
PE	Physical exercise
UPF	Ultra-processed food
WFPB	Whole food plant-based
WFSBP	World federation of societies for biological psychiatry

Supplementary Information

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Supplementary Material 1

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Consent for publication

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Competing interests

The authors declare no competing interests.

Author details

¹Rehabilitation Sciences Programs, Department of Occupational Science and Occupational Therapy, Department of Physical Therapy, Faculty of Medicine, University of British Columbia, Vancouver, Canada

²Department of Physical Therapy, Faculty of Medicine, University of British Columbia, Vancouver, BC, Canada

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